

OrthoHMM: HMM-Centered Group Inference With Phylogenetic Refinement

Condensed scientific draft, updated 2 October 2026. Not submission-ready. The [extended manuscript](#) retains detailed methods, historical analyses, citations and audit records. This main text does not supersede frozen protocols or historical result manifests.

Abstract

Orthology inference requires balancing homolog recovery against separation of paralogs. We evaluated an HMM-centered pipeline with candidate-family expansion and phylogenetic refinement against established orthology tools. QfO and OrthoBench were the primary development-exposed benchmarks; a frozen YGOB evaluation assessed transfer to additional taxa. Phylogenetic OrthoHMM improved on its high-sensitivity configuration but did not consistently outperform full OrthoFinder. On OrthoBench, their group-recovery F1 values were 74.11% and 72.74%, respectively. On corrected QfO, OrthoFinder had higher VGNC, SwissTrees and TreeFam-A point estimates, whereas OrthoHMM had higher GO/EC similarity and FAS. A matched-recall simulation control supported an initial HMM-search contribution in a fixed graph procedure, not full-pipeline superiority. Independent-family generalization, uncertainty for several QfO endpoints and controlled comparative timing remain unresolved. The evidence supports a bounded contribution and an explicit precision-recall trade-off, not universal accuracy or efficiency claims. Synthetic null scoring revealed composition-dependent behavior of the approximate significance filter.

Methods

The retained configurations are high-sensitivity OrthoHMM and the `satellite_v2` phylogenetic pipeline. The original OrthoHMM preprint (Steenwyk et al. 2024) describes its lineage, not all subsequent implementation changes. The prospective method was frozen at `7f3a9e4`, with BLOSUM62, E-value threshold $1e-4$, Leiden CPM resolution 0.1 and seed 4 (Traag, Waltman, and van Eck 2019; Traag, Van Dooren, and Nesterov 2011). The phylogenetic configuration expands candidate families, infers gene and species trees, and applies positive-paralogy pair inference. Historical runs retain their actual revisions rather than inheriting this prospective pin. The [method diagram](#) distinguishes initial search, profile refinement, candidate expansion and phylogenetic inference.

Initial search uses banded local maximum-path scoring with match, insert and delete states, integer BLOSUM62 emissions and fixed transition costs. Raw scores receive an approximate significance filter before geometric-mean length normalization. MSA-derived profiles learn match emissions, while retaining uniform transition costs; strict expansion requires both a member-score threshold and an initial sequence-supported anchor. This is not demonstrated HMMER/phmmer equivalence or calibrated assignment confidence. The [numerical specification](#) records the recurrence, constants, profile construction and acceptance rules at the full frozen source revision. No scoring/default change accompanies it.

A prespecified null-score protocol examined that significance filter without fitting parameters. Three residue compositions and lengths 50/150/400 each used ten seeds and 1,000 independently drawn query-target pairs per seed. Each pair had a one-target database and was scored with full and width-64 bands: 90,000 independent pairs, not 180,000 independent observations. Five fixed cutoffs yielded 90 tail endpoints with Bonferroni-adjusted exact binomial intervals. These intervals describe the synthetic generator, not reference-family or orthology uncertainty.

Comparators were OrthoFinder 3.1.5 (David M. Emms et al. 2026a, 2026b), SonicParanoid 2.0.9 (Cosentino, Sriswasdi, and Iwasaki 2024), ProteinOrtho 6.3.6 (Klemm, Stadler, and Lechner 2023), FastOMA 0.3.5 (Majidian et al. 2025) and OrthoMCL 1.4 (Li, Stoeckert, and Roos 2003). These method references do not replace run-specific executable provenance. OrthoFinder's sequence-only MCL checkpoint was a diagnostic output, not a separately finalized phylogenetic analysis. FastOMA used a supplied OrthoFinder species tree. QfO inputs included native ortholog pairs, native post-clustering relations or group-derived cross-species pairs as appropriate; these are not interchangeable output semantics. The generated comparison reports each conversion and prediction count.

OrthoBench measures curated group recovery (David M. Emms and Kelly 2020). QfO (Quest for Orthologs consortium et al. 2016; Altenhoff et al. 2020) reports GO and EC similarity, VGNC, SwissTrees and TreeFam-A F1, and FAS separately. Their arithmetic mean is a project-defined secondary summary, not an official QfO F1. Three Kingdoms is supplementary BUSCO-reference recovery (Manni et al. 2021), not genome-wide orthology truth. QfO and OrthoBench were repeatedly inspected during development. The YGOB protocol, using the curated homology/syteny resource (Byrne and Wolfe 2005), froze evaluation before test-score inspection; overlap assessment limits its interpretation to novel-taxon transfer rather than family-disjoint validation.

Uncertainty analyses resample declared reference units and recompute the benchmark statistic within replicates. They retain prespecified multiplicity adjustments, negative findings and unavailable contrasts. Curated families are not automatically independent: shared evolutionary history and predictions joining families can violate exchangeability. Dependent protein pairs are not treated as independent observations to obtain narrower intervals.

The prespecified QfO parameter neighborhood retained seven arms: the frozen control and six one-at-a-time changes. CPM resolutions were 0.08/0.12 versus 0.1, candidate minimum normalization 0.024/0.036 versus 0.03, and minimum margin 1.2/1.8 versus 1.5. SwissTrees F1 was recomputed from macro precision and recall in each of 100,000 shared family-bootstrap draws, not averaged across family F1 values. The seed was 20260925; Bonferroni adjustment retained all 18 planned endpoints across six contrasts and three metrics. This is development-exposed local sensitivity, not independent method selection. Frozen parameter protocol.

Results

Accuracy Depends On The Endpoint

Phylogenetic OrthoHMM achieved 74.1061% OrthoBench F1 versus 72.7365% for full OrthoFinder. In the exploratory eight-method comparison, the F1 difference was +1.370 percentage points (21-endpoint adjusted interval [-7.281, 12.166]). The precision difference was +15.705 [1.175, 30.634] and recall difference -13.151 [-26.947, -0.609]. This is a precision-recall tradeoff, not established F1 superiority. The complete panel uses 100,000 paired RefOG draws and remains conditional on family exchangeability and development exposure. All 21 contrasts. The complete interval figure shows F1, precision and recall

on the same difference scale without ranking methods by their observed effects. A fresh installed full run reproduced all 59,770 OrthoHMM groups exactly; this establishes reproducibility, not independent accuracy or a controlled runtime comparison. Full-run verification. A second full run using a separately reconstructed base Python and fresh inference/reader environments also reproduced all groups and all 70 family score records. Native pair, confidence, reconciliation-event and hierarchy files were byte-identical to the earlier run. This extends same-host environment reproduction, not biological validation or complete archive restoration. Reconstructed-base verification. A third full run used restored local execution assets and separately acquired upstream inputs. Its independent scientific admission reproduced all 59,770 groups, all 70 family score records and four native TSVs exactly, without reusing phylogenetic checkpoints. This validates local archive-to-results reproduction, not cross-host/OS restoration, independent accuracy or controlled timing. Restored-archive verification.

On corrected QfO, phylogenetic OrthoHMM versus full OrthoFinder scored 0.901690 versus 0.988546 on VGNC, 0.833513 versus 0.848413 on SwissTrees, and 0.614864 versus 0.791918 on TreeFam-A. OrthoHMM had higher GO similarity (0.490349 versus 0.469548), EC similarity (0.965650 versus 0.936130) and FAS (0.762993 versus 0.691422). These endpoints differ in reference scope and eligibility; they do not form a single accuracy ranking. All eight methods.

An exploratory VGNC deletion diagnostic removed each of 16,844 reference blocks and its incident scored pairs in turn, across all eight methods. Every comparator retained a negative F1 difference against full OrthoFinder for every deletion. For phylogenetic OrthoHMM the range was [-8.7353, -8.6729] percentage points. These are sensitivity ranges, not confidence intervals: dependent deletions change the fixed scored table, and single-block sign stability does not establish population or joint-deletion robustness. Native inference and eligibility were not rerun.

For corrected SwissTrees, phylogenetic-minus-full-OrthoFinder F1 was -1.4900 percentage points, with a multiplicity-adjusted interval of [-8.7078, 7.2090]. Phylogenetic-minus-high-sensitivity OrthoHMM was +14.8015 points [4.9791, 26.6015]. These conditional estimates use 18 development-exposed families and a 24-endpoint adjustment; inclusion of zero is not equivalence. Candidate expansion also changes between OrthoHMM configurations, so their contrast does not isolate reconciliation. Protocol and contrast evidence.

All 56 method-pair GO/EC comparisons had identical six-decimal scores on shared scored pairs. Aggregate differences arose from different eligible pair sets and their denominators. Restricting evaluation to intersections would change the endpoint rather than resolve uncertainty. Historical raw hashes were verified for all 16 input tables. Scored-pair audit. The complete pair-composition figure displays the shared-denominator and exclusive-pair terms for all seven comparators against full OrthoFinder. Phylogenetic OrthoHMM's shared pairs account for 90.12% of its GO scored set but 46.40% of OrthoFinder's, and 86.59% versus 58.00% for EC. The original rounded-mean differences, +2.080130 and +2.952029 score points, are the sums of much larger opposing terms. These fractions describe eligible scored pairs, not proteome-wide coverage; the decomposition is arithmetic, not causal attribution or paired uncertainty.

Component Evidence Is Bounded

The corrected QfO factorial supported candidate-expansion-by-reconciliation interactions on SwissTrees F1, while all four profile-refinement F1 intervals included zero. Reconciliation increased precision and reduced recall. These results support conditional component effects, not proof that profile expansion universally improves orthology inference. Complete factorial.

In 35 matched-recall simulation datasets spanning seven conditions and five seeds, HMM-derived search evidence yielded mean downstream graph F1 of 83.6097%, versus 80.5180% for DIAMOND (Buchfink, Reuter, and Drost 2021). The adjusted paired seed-block interval for the +3.0917-point difference was [1.6782, 4.5313]. Profile expansion and phylogeny were off. This is a development-exposed fixed-graph comparison, not a comparison against OrthoFinder or a matched-effort result. Score rankings and hit identities differ, so the design does not isolate a causal mechanism. [Matched-recall control](#).

Synthetic Null Tails Depend On Composition

At the frozen $E < 1e-4$ cutoff, width-64 scoring passed 0/1/0 of 10,000 ordinary-background pairs at lengths 50/150/400. These sparse counts do not establish rare-tail calibration. Half-background-plus-half-glutamine sampling passed 89.83%, 100% and 100%; its adjusted intervals were [88.7484%, 90.8453%] and [99.9181%, 100%] for each longer length. All [90 endpoints and their figure](#) are retained. A hit here is only a forced pair passing the approximate significance gate, not a predicted ortholog or an observed pipeline false positive. No prefilter or biological inference ran. The strong synthetic composition is not a measured real-proteome prevalence. The [audit](#) reproduces all endpoint counts and 180 sparse reference-Python scores, not all native scores. No coefficients, thresholds or defaults were fitted or promoted.

Local Parameter Sensitivity Did Not Establish Improvement

The [complete seven-arm panel](#) includes separately admitted private-runtime high-CPM recovery. Its SwissTrees F1 was 83.253016% versus 83.351322% for the frozen control, a difference of -0.098306 percentage points with adjusted interval [-1.050447, 0.747525]. All 18 adjusted intervals include zero; no statistically supported parameter improvement is established. Identical observed normalization-arm statistics do not imply identical whole-proteome predictions or equivalence on unseen families. No defaults were changed. The [all-arm figure](#) retains every contrast and the full adjustment denominator.

Historical SIGSEGV, allocator and admission failures remain failed. Recovery followed a validated content-equivalent private control and has [separate inference](#), [conversion and score evidence](#); it neither repairs those earlier attempts nor proves their cause or memory safety. This panel is not an OrthoFinder superiority test or a controlled runtime comparison.

Transfer And Biological Recovery Reveal Trade-Offs

Frozen YGOB group-recovery F1 was 92.233654% for phylogenetic OrthoHMM and 92.318524% for full OrthoFinder. Their difference was -0.084870 points, with adjusted interval [-0.622528, 0.445222]. OrthoHMM had higher precision and lower recall. The evaluation projects predictions onto the reference universe and assumes exchangeable pillars; it does not establish unrestricted generalization or resolved pairwise orthology. [YGOB results](#).

A descriptive partition by the frozen overlap screen retained all original false-positive allocations. In screen-negative pillars, phylogenetic OrthoHMM versus full OrthoFinder had F1 of 57.93% versus 45.02%, precision of 55.92% versus 30.74%, and recall of 60.10% versus 84.03%. In screen-positive pillars, their F1 values were 93.37% versus 95.01%. The negative stratum contained 2,893 singleton pillars out of 3,298, compared with 2,017 out of 6,952 in the positive stratum. This composition difference and the screen's inability to exclude remote homology prevent independent-family or causal interpretations. The [all-method stratum figure](#) is descriptive, with no new uncertainty estimate or method tuning.

The prespecified whole-genome-duplicate application used experimental evidence from Kuzmin and colleagues (Kuzmin et al. 2020). All 240 experimental pairs were retained, with 239 input-eligible and 231 shared-reference-pillar pairs. On this development-exposed application, phylogenetic OrthoHMM separated 238 of 239 input-eligible pairs, versus 58 for high sensitivity and 236 for full OrthoFinder. Among the 231 reference-eligible pairs, however, only 193 OrthoHMM separations retained at least one non-*S. cerevisiae* reference homolog with each anchor, versus 56 for high sensitivity and 227 for full OrthoFinder. Mean per-pair homolog coverage was 82.338%, 99.149% and 98.413%, respectively. Coverage counts reference homologs in the union of the anchor groups and can be high even when the anchors are merged; it is not orthology recall. The [complete five-method comparison](#) also retains SonicParanoid and the diagnostic OrthoFinder MCL checkpoint.

Phylogenetic OrthoHMM minus full OrthoFinder had a supported-separation difference of -14.719 percentage points, with adjusted interval [-21.645, -8.225], and a coverage difference of -16.075 points [-19.755, -12.496]. These exploratory percentile intervals use 20,000 paired pillar resamples and a 12-endpoint adjustment. Each eligible pair occupies a distinct pillar; these conditional estimates do not establish independent generalization. Supported separation does not establish cross-species copy-specific orthology. In the [six prespecified case traces](#), five focal homologs left their anchor groups during root-lineage reconstruction, before satellite constraints. They remained in other output groups; no example was replaced to improve the result.

A subsequent [fixed-tree diagnostic](#) evaluated all four existing root-duplication rules on the same seven candidate families and six examples, with its protocol frozen before alternative outcomes were examined. The supported-children and confidence rules reproduced the species-overlap baseline partitions exactly. The mapped-event rule reduced homolog coverage in four of five reference-eligible examples, without improving supported separation. None recovered the five focal homologs into either anchor group. Native pair predictions and confidence annotations remained unchanged. This post hoc intervention rejects these three alternatives as a repair on the fixed inputs, not as methods in general. It neither establishes topology error nor identifies ancestral-copy truth, and no defaults were changed.

Discussion And Limitations

The supported contribution is an HMM-centered alternative with measurable component effects and explicit accuracy trade-offs. Initial HMM search has bounded support in matched-recall simulations; additional profile refinement has not demonstrated a general benefit. Neither simulation evidence nor an OrthoBench point advantage establishes superiority over full OrthoFinder.

The approximate significance formula is not universally calibrated under the tested synthetic conditions. The null experiment does not measure real-data orthology false-positive rates or causally explain any benchmark difference; independent calibration would be needed before claiming statistical significance for arbitrary compositions, search spaces, profiles or band settings.

Original TreeFam-A family mappings and complete source trees remain unavailable. Public QfO container searches now cover recognized benchmark build contexts for [all 40 listed tag names](#) at retained digests, without finding original NHX/mapping filename candidates. Other layers and archive-embedded or differently named contents remain outside that search; no family labels are inferred from its negative result. Public archive recovery yielded historical Selectome subtrees, but all are restricted to Euteleostomi. A [taxonomic coverage audit](#) found that 55,933 of 79,320 retained reference relations (70.52%) have at least one endpoint outside that clade. Those relations cannot be reconstructed from these subtrees under a species-consistent mapping; the other 23,387 relations are only potentially in scope, not demonstrated reconstructions. This is a coverage exclusion, not an accuracy effect or a basis for family-

level intervals. VGNC dependence and rare-error diagnostics do not justify the candidate confidence-interval procedure. Valid paired uncertainty for GO/EC, FAS and the secondary mean is also unfinished. These gaps must remain visible alongside point estimates. The corrected FAS sample audit reproduces all eight means, but scored fractions range from 0.0067% to 58.46% of reported eligible pairs and every sample reuses proteins across pairs. Native pair-level standard errors do not resolve this comparison uncertainty. The requested-score audit found 1-49 missing new scores per method except OrthoMCL, which lacked 1,252 of 9,000. Assuming omitted scores lie in [0,1], its intended-sample mean is bounded by 0.724872-0.736975; this is not a confidence interval. A native-complexity audit found no above-limit protein in saved new-score pairs, but found them in precomputed pairs. This agrees with the demonstrated cutoff mechanism without identifying historical omitted pairs or establishing comparison bias. The completed eligible-population audit matches all eight methods' native logged counts, reusing six fixed recounts and freshly scanning two. Pairs absent from the precomputed lookup constitute 5.13% of satellite_v2's eligible set, 16.86% of full OrthoFinder's and 82.05% of its sequence-only checkpoint's. Assuming their scores lie in [0,1] gives conservative full-mean bounds of 0.733966-0.785282 for satellite_v2 and 0.605946-0.774511 for full OrthoFinder. These overlapping bounds neither establish a full-eligible-set advantage nor replace native sample scores or confidence intervals. Some lookup-missing pairs were newly scored in native samples; those values are not used to tighten these bounds. An eight-method stratum-weight decomposition separates count rounding from omission-induced changes in the saved mixture. Holding the observed stratum means fixed, native-minus-reweighted-saved-strata differences are +0.001318947 FAS units for OrthoMCL and +0.000030716 for phylogenetic OrthoHMM. These are numerical diagnostics, not corrected benchmark scores or estimates of population bias. Omissions can also change the stratum means; the unknown omitted pair identities, unseeded selection and dependence remain unresolved. No FAS confidence interval or new method ranking follows. The completed simulations and tree perturbations do not cover arbitrary evolutionary conditions, and novel-taxon YGOB testing retains homolog-family overlap with development data.

Recovered OrthoMCL results retain sequence-specific BLAST failures. The separately recovered high-CPM result does not replace failed historical runs; no missing score was imputed and no failure was recategorized as a success. Shared-host runs and the historical DGX panel are descriptive resource records; the replacement controlled Threadripper timing panel has not run. No controlled speedup is claimed. Versioned archival release, complete dependency/data redistribution review, journal formatting and final visual review remain open.

Reproducibility And Availability

The extended manuscript links frozen protocols, source revisions, seeds, input/output manifests, conversion audits, executable workflows and generated figures. Installed inference and independent readback have been validated on full OrthoBench, but that is not proof of all-method cross-host portability. Historical locks are retained as provenance; patched replacement environments must be distinguished from the binaries used for original scores. The QfO parameter numerical component passed fresh local archive extraction, copied byte verification and reproduction of all 18 endpoints within 1e-12 using unchanged arithmetic. A Python-event guard rejected its original-checkout canary and observed no later original-path events or project-module imports. This is not OS containment, a new independent statistical implementation, raw-reference recount, native inference restoration, cross-host validation or redistribution clearance.

A separate descriptive-table component restored under standard-library Python 3.12 and reproduced all 208 rows and 984 score/difference cells across four SwissTrees tables and eight methods. This verifies arithmetic from retained sufficient statistics and annotations, not raw-source admission, new benchmark

scoring or bootstrap uncertainty. An input-only private archive workflow then restored the complete duplication and fragment dependency panels, retaining all 11 and 1,774 original record occurrences respectively. Independently pinned archive and binding digests preserve source identities at new locations. All 27 affected regression/options cases passed using the restored inputs, including four raw-export regressions with unchanged assertions. The fresh children blocked original-checkout reads and subprocesses; these Python-event guards are not OS containment. Native annotation extraction/admission was not rerun, and private archives remain unuploaded with redistribution uncleared. Neither component establishes complete executable study restoration or all-method cross-host portability.

The null-score observations retain all 180,000 synthetic score evaluations and their source/runtime pins. Its public numerical readback checks all 90 tail endpoints without loading the native kernel; this is not a complete inference runtime archive.

The progress ledger records completed work and unmet requirements. No submission-ready release or archival DOI is claimed. Dated main-text HTML/PDF and review archives preserve the exact inputs of their own snapshots; older versions are not rendered copies of revised Markdown. A refreshed review is not a complete executable study release. Final manuscript/figure/archive reconciliation after outstanding analyses remains required.

References

The bibliography attributes the methods and resources cited above. Full execution evidence and additional dependency references remain in the extended manuscript; this selected list is not complete software attribution.

- Altenhoff, Adrian M, Javier Garrayo-Ventas, Salvatore Cosentino, David Emms, Natasha M Glover, Ana Hernández-Plaza, Yannis Nevers, et al. 2020. “The Quest for Orthologs Benchmark Service and Consensus Calls in 2020.” *Nucleic Acids Research* 48 (W1): W538–45. <https://doi.org/10.1093/nar/gkaa308>.
- Buchfink, Benjamin, Klaus Reuter, and Hajk-Georg Drost. 2021. “Sensitive Protein Alignments at Tree-of-Life Scale Using DIAMOND.” *Nature Methods* 18 (4): 366–68. <https://doi.org/10.1038/s41592-021-01101-x>.
- Byrne, Kevin P., and Kenneth H. Wolfe. 2005. “The Yeast Gene Order Browser: Combining Curated Homology and Syntenic Context Reveals Gene Fate in Polyploid Species.” *Genome Research* 15 (10): 1456–61. <https://doi.org/10.1101/gr.3672305>.
- Cosentino, Salvatore, Sira Sriswasdi, and Wataru Iwasaki. 2024. “SonicParanoid2: Fast, Accurate, and Comprehensive Orthology Inference with Machine Learning and Language Models.” *Genome Biology* 25 (1). <https://doi.org/10.1186/s13059-024-03298-4>.
- Emms, David M, and Steven Kelly. 2020. “Benchmarking Orthogroup Inference Accuracy: Revisiting Orthobench.” *Genome Biology and Evolution* 12 (12): 2258–66. <https://doi.org/10.1093/gbe/evaa211>.
- Emms, David M., Yi Liu, Laurence Belcher, Jonathan Holmes, and Steven Kelly. 2026a. “OrthoFinder: Improved Phylogenetic Orthology Inference with Enhanced Accuracy and Scalability.” *Nature Methods* 23 (7): 1327–33. <https://doi.org/10.1038/s41592-026-03126-6>.
- . 2026b. “Author Correction: OrthoFinder: Improved Phylogenetic Orthology Inference with Enhanced Accuracy and Scalability.” *Nature Methods*, August. <https://doi.org/10.1038/s41592-026-03238-z>.
- Klemm, Paul, Peter F. Stadler, and Marcus Lechner. 2023. “Proteinortho6: Pseudo-Reciprocal Best Alignment Heuristic for Graph-Based Detection of (Co-)orthologs.” *Frontiers in Bioinformatics*

- 3 (December). <https://doi.org/10.3389/fbinf.2023.1322477>.
- Kuzmin, Elena, Benjamin VanderSluis, Alex N. Nguyen Ba, Wen Wang, Elizabeth N. Koch, Matej Usaj, Anton Khmelinskii, et al. 2020. "Exploring Whole-Genome Duplicate Gene Retention with Complex Genetic Interaction Analysis." *Science* 368 (6498). <https://doi.org/10.1126/science.aaz5667>.
- Li, Li, Christian J. Stoeckert Jr., and David S. Roos. 2003. "OrthoMCL: Identification of Ortholog Groups for Eukaryotic Genomes." *Genome Research* 13 (9): 2178–89. <https://doi.org/10.1101/gr.1224503>.
- Majidian, Sina, Yannis Nevers, Ali Yazdizadeh Kharrazi, Alex Warwick Vesztröcy, Stefano Pascarelli, David Moi, Natasha Glover, Adrian M. Altenhoff, and Christophe Dessimoz. 2025. "Orthology Inference at Scale with FastOMA." *Nature Methods* 22 (2): 269–72. <https://doi.org/10.1038/s41592-024-02552-8>.
- Manni, Mosè, Matthew R Berkeley, Mathieu Seppey, Felipe A Simão, and Evgeny M Zdobnov. 2021. "BUSCO Update: Novel and Streamlined Workflows Along with Broader and Deeper Phylogenetic Coverage for Scoring of Eukaryotic, Prokaryotic, and Viral Genomes." *Molecular Biology and Evolution* 38 (10): 4647–54. <https://doi.org/10.1093/molbev/msab199>.
- Quest for Orthologs consortium, Adrian M Altenhoff, Brigitte Boeckmann, Salvador Capella-Gutierrez, Daniel A Dalquen, Todd DeLuca, Kristoffer Forslund, et al. 2016. "Standardized Benchmarking in the Quest for Orthologs." *Nature Methods* 13 (5): 425–30. <https://doi.org/10.1038/nmeth.3830>.
- Steenwyk, Jacob L, Thomas J. Buida III, Antonis Rokas, and Nicole King. 2024. "OrthoHMM: Improved Inference of Ortholog Groups Using Hidden Markov Models." Preprint. openRxiv. <https://doi.org/10.1101/2024.12.07.627370>.
- Traag, V. A., P. Van Dooren, and Y. Nesterov. 2011. "Narrow Scope for Resolution-Limit-Free Community Detection." *Physical Review E* 84 (1). <https://doi.org/10.1103/PhysRevE.84.016114>.
- Traag, V. A., L. Waltman, and N. J. van Eck. 2019. "From Louvain to Leiden: Guaranteeing Well-Connected Communities." *Scientific Reports* 9 (1). <https://doi.org/10.1038/s41598-019-41695-z>.